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8873/02

24 August 2021

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Civics Group, centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
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This document consists of **27** printed pages and **1** blank page.

8873 / TJC Prelim / 2021

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Section A

Answer **all** the questions in the spaces provided.

- 1 (a) Chlorine reacts with iodine to form a compound **T**, ICl_7 . When dissolved in an excess of aqueous potassium iodide, **T** liberates iodine, I_2 , which is the only iodine-containing product in the reaction.

(i) State the oxidation number of iodine in ICl_7 . [1]

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(ii) Write the equation for the reaction between **T** and potassium iodide. [1]

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(iii) Calculate the amount of iodine liberated when 1.00 g of **T** reacts with an excess of aqueous potassium iodide. [1]

- (b) ICl_3 can also be formed from iodine and chlorine.

(i) Draw the dot-and-cross diagram for ICl_3 . [1]

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- (ii) With reference to its molecular shape and partial charges, explain if ICl_3 is a polar molecule. [2]

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- (c) Both chlorine and iodine can be formed from the decomposition of its respective hydrides. State and explain the trend in the thermal stability of the Group 17 hydrides going down the group. [2]

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- (d) Sodium thiosulfate is a common reagent used for the reaction with iodine. Write a balanced equation for the reaction between sodium thiosulfate and iodine and calculate the volume of 1.00 mol dm^{-3} sodium thiosulfate, in cm^3 , required to react with all the iodine liberated in (a)(iii). [2]

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[Total: 10]

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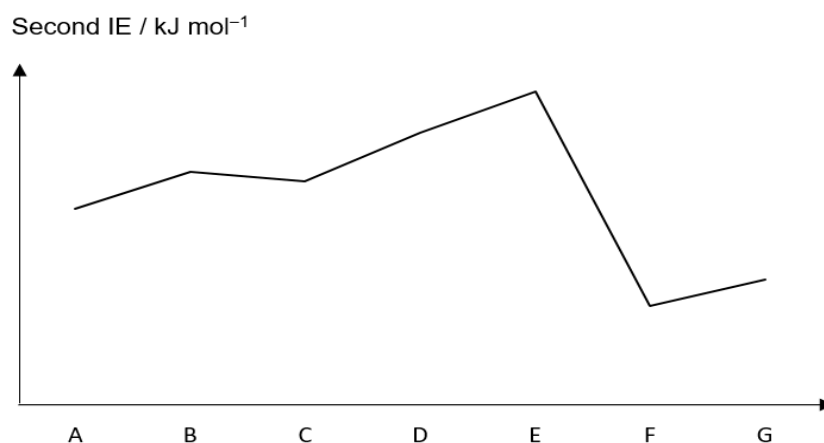
- (a) Excited states can be studied to gain information about the energies of orbitals that are unoccupied in an atom's ground state. The excited state of an element, **Q**, is represented by the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2 4p^1$.

- (i) Identify the element **Q**. [1]

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- (ii) Draw the energy level diagram showing the electronic configuration of element **Q** in its ground state. [2]

- (b) The second ionisation energies of seven consecutive elements **A** to **G** in the Periodic Table are shown below.



- (i) Define, with the aid of an equation, the second ionisation energy of oxygen. [2]

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- (ii) Explain the discontinuity in second ionisation energies between **E** and **F**, and between **B** and **C**. Hence deduce which element **A** to **G** is oxygen. [3]

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- (c) (i) State and explain the variation in bonding in Period 3 oxides in terms of electronegativity. [2]

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W, **X**, **Y**, and **Z** are four consecutive elements in the fourth period of the Periodic Table. The letters are not the actual symbols of the elements.

W forms an oxide that reacts with both acids and bases.

Z is a solid that can exist as several different allotropes. **Z** burns in air to form ZO_2 which dissolves in water to form an acidic solution. This solution reacts with sodium hydroxide to form the salt Na_2ZO_3 .

(ii) Suggest the identities of **W** and **Z**. [1]

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(iii) Write equations for the reactions of oxide of **W** with sodium hydroxide and hydrochloric acid respectively. [2]

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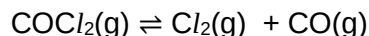
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- 3 Chlorine is produced together with carbon monoxide when phosgene, COCl_2 , undergoes dissociation according to the equation below:



The above reaction takes place in a 2 dm^3 reaction vessel. At the start of the reaction, there was 4.0 mol of phosgene in the vessel. When dynamic equilibrium was established, only 1.6 mol of phosgene was left.

- (a) Write the K_c expression for the above reaction, stating its units. [1]
- (b) Calculate the equilibrium amounts of CO and Cl_2 , and hence, determine the value of K_c for the above reaction, showing your working clearly. [2]

- (c) State *Le Chatelier's Principle*. [1]

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- (d) State and explain how the equilibrium composition might change if the above reaction is subjected to a decrease in pressure. [2]

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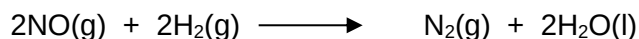
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- 4 (a) Nitrogen monoxide, NO(g), is a colourless, toxic gas that is formed by the oxidation of nitrogen. It can be reduced by hydrogen, H₂(g), under certain conditions to form harmless products.



- (i) Define the term 'rate of reaction'. [1]

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- (ii) Identify a change in the reaction mixture that would allow the rate of reaction to be measured. [1]

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A series of experiments was conducted to determine the rate equation for the above reaction. The following data was obtained.

Experiment	Initial [NO] / mol dm ⁻³	Initial [H ₂] / mol dm ⁻³	Initial Rate / mol dm ⁻³ s ⁻¹
1	2.50 × 10 ⁻³	2.50 × 10 ⁻³	1.27 × 10 ⁻³
2	2.50 × 10 ⁻³	4.60 × 10 ⁻³	2.34 × 10 ⁻³
3	5.00 × 10 ⁻⁴	7.50 × 10 ⁻³	1.52 × 10 ⁻⁴

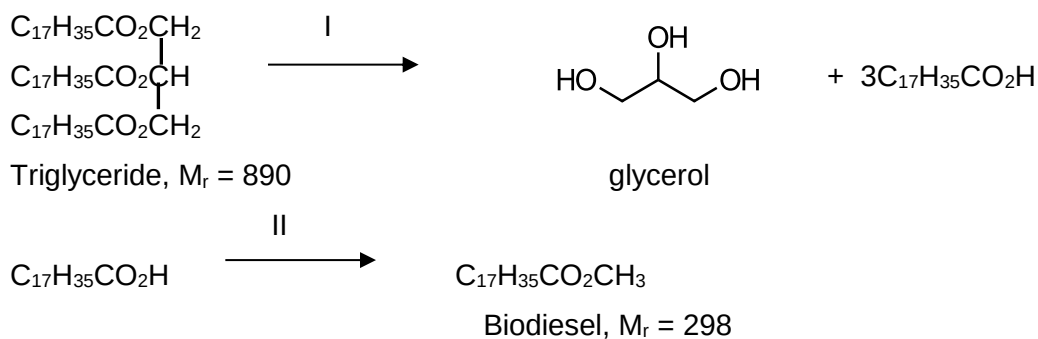
- (iii) Determine the order of reaction with respect to NO and H₂. Hence state the rate equation. [3]

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- (iv) Using the data from Experiment 2, calculate the rate constant, k , stating its units clearly. [2]

- (b) Recently much interest has been shown in the production of the fuel biodiesel from algae. 50% of the mass of the dried algae are triglycerides. To convert triglycerides into biodiesel, the following processes are carried out.



- (i) State the reactants and conditions for reactions I and II. [2]

Reaction I :

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Reaction II :

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- (ii) Calculate the mass of biodiesel that can be produced from 1000 kg of dried algae. [2]

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- (iii) The production of biodiesel is at present an expensive process.

Suggest a reason why the development of biodiesel as an alternative to fossil fuels is important. [1]

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- (c) Oxides of nitrogen are often produced in the exhaust fumes of car engines, together with other pollutants including hydrocarbons and carbon monoxide.

- (i) Suggest another negative environmental consequence of the oxides of nitrogen. [1]

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- (ii) Catalytic converters are used to reduce the emission of these harmful pollutants.

Write an equation for the reaction that occurs in the catalytic converter that removes both carbon monoxide and nitrogen monoxide from the exhaust gases. [1]

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- (iii) Catalytic converters contain platinum nanoparticles.

Explain why platinum nanoparticles are used in the catalysis. [1]

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- (d) Human blood contains a buffer of carbonic acid, H_2CO_3 , and bicarbonate anion, HCO_3^- , in order to maintain blood pH between 7.35 and 7.45. A value higher than 7.8 or lower than 6.8 can lead to death.

Explain, with the aid of equations, the role of carbonic acid and bicarbonate ion in controlling pH in blood. [3]

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- 5 (a) Short chain alkenes is a feedstock for polymer and they can be obtained from the cracking of hydrocarbon chains.

Catalytic cracking is a process where long hydrocarbon chains are broken down into mixtures of molecules with shorter chains.

When an alkane $C_{11}H_{24}$, was subjected to catalytic cracking, three products were identified. The molecular mass of the mixture was 28, 42 and 44 with the relative proportions of 1: 2: 1 respectively. One of the products does not decolourise bromine.

Write a balanced equation for the catalytic cracking of $C_{11}H_{24}$. Show clearly the structural formulae of the three products and indicate the product that does not decolourise bromine. [3]

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Nylon and polypropene are polymers that have a wide range of uses.

- (b) Apart from making containers, polypropene, also known as polypropylene, and nylon are used in making material for single use face masks. Polypropene is used for the mask layers while nylon is used for the ear loops of masks.

According to WHO mask guide, masks are advised to have an inner most layer of absorbent, hydrophilic material, an outermost layer of hydrophobic material and a middle hydrophobic layer to enhance filtration or retain droplets. Masks should also meet the minimum requirements of filtration, breathability and fit.

Masks can also be made of woven or nonwoven fabric. The difference between the two fabrics is as shown in the table below.

	Nonwoven fabric	Woven fabric
Bonded by	Mechanical, glue or use of heat to melt fibers. Randomly oriented fibers are compressed.	Weaving yarns together via different construction patterns.
Properties	Light and cheap.	Washable and Durable. Not stretchable unless elastic fibers are used.

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- (i) Suggest why the inner layer of the masks needs to be hydrophilic with high absorbance while the outer layer of the mask needs to be hydrophobic. [1]

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- (ii) With the knowledge of the bonding in polypropene, explain the suitability of using polypropene as the outer layer of single use masks [1]

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- (iii) Suggest two reasons why nonwoven polypropene is suitable in the manufacture of single use face mask. [2]

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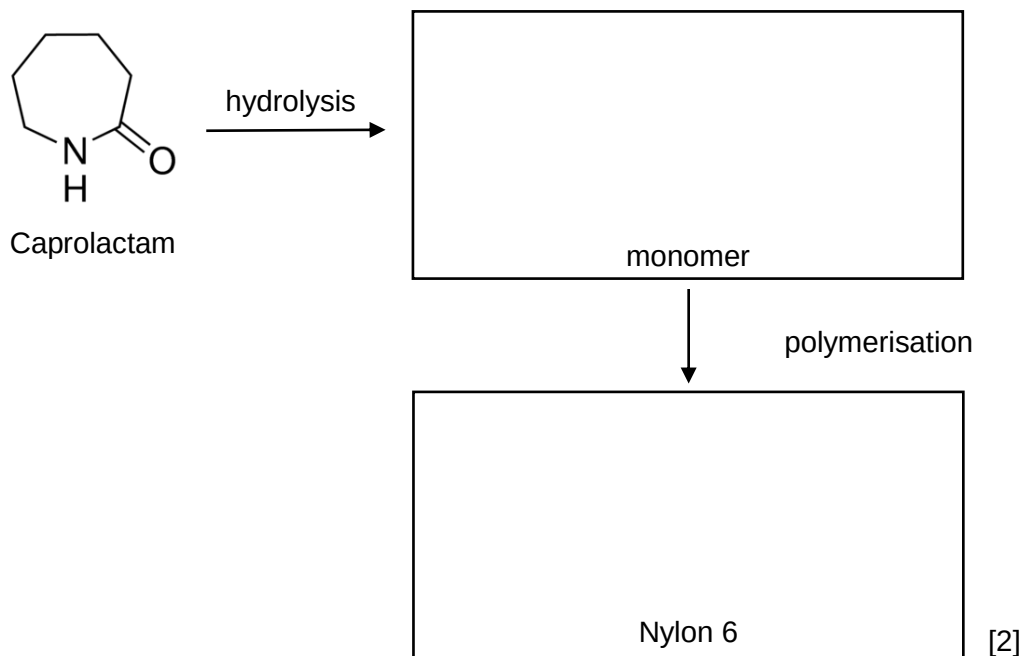
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- (c) Nylon 6 is made via ring opening polymerisation of caprolactam. The reaction is initiated by the hydrolysis of Caprolactam and the subsequent polymerisation of the monomer.

- (i) Draw the structure of the monomer and the polymer, Nylon 6, in the scheme below.



Some nylon fabric can be prone to creasing.

In order to remove creases in a nylon garment, an online blog proposed the following steps.

Step 1: Wet your wrinkled nylon garment thoroughly in lukewarm water. Wring out excess water.

Step 2: Place the wet nylon garment in the dryer. Turn the dryer to permanent press setting for synthetics (approximately 40°C) to prevent creases from forming during the drying process.

Step 3: Hang the nylon garment in a dry spot.

Step 4: Use a low heat iron to iron the nylon garment. Do not use high heat.

- (ii) Suggest why wetting the nylon garment in Step 1 is necessary in removing creases in the nylon garment. [1]

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- (iii) Suggest why the use of low heat iron will remove creases in nylon garments. [1]

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- (iv) Suggest why high heat iron may not be suitable in removing creases. [1]

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- (v) Nylon polymers can be mixed with a wide variety of additives to achieve many different property variations. One method is the use of radiation to form cross links in Nylon.

Suggest how the use of additives to form cross-links changes the property of Nylon. [1]

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Section B

Answer **one** question from this section in the spaces provided.

- 6 (a) Sodium chloride and iodine have lattice structures.

Describe the lattice structures and bonding in iodine and sodium chloride.

You may draw diagrams to illustrate your answer.

[4]

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- (b) (i) With the aid of an equation, define *lattice energy of sodium oxide*.

[1]

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- (ii) The lattice energy of sodium oxide is $-2720 \text{ kJ mol}^{-1}$.

Suggest whether sodium oxide or sodium chloride would have the more exothermic lattice energy.

[2]

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- (iii) Lithium reacts less readily in air than sodium.

With reference to the Data Booklet, explain why lithium is less reactive than sodium.

[2]

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- (c) Hydrogen peroxide is an eco-friendly substitute for chlorinated agents in industrial whitening of cotton. It breaks down to water and oxygen, which are not harmful to the environment.



This reaction can be speeded up by adding some silver powder.

- (i) In the same diagram, construct a reaction pathway diagram for the decomposition of hydrogen peroxide with and without silver. [2]

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- (ii) An enzyme, catalase, can be used to speed up the decomposition of hydrogen peroxide.

Papain is an enzyme that cleaves fibrous protein.

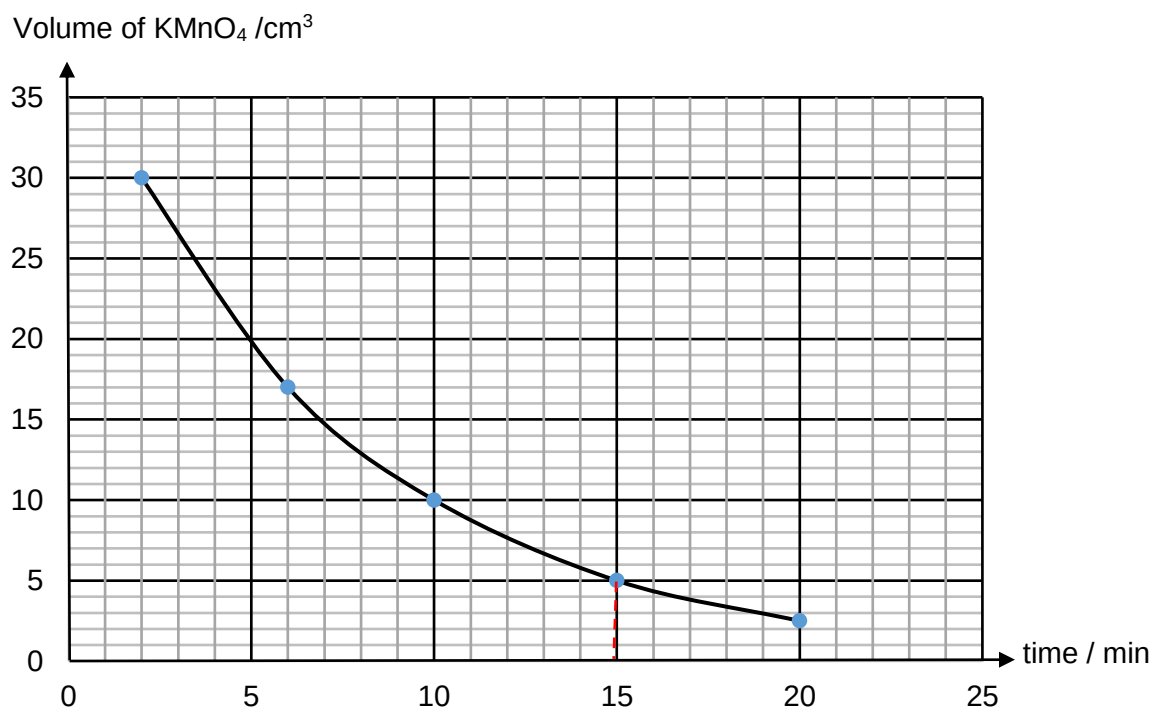
Explain the effect on the rate of decomposition of hydrogen peroxide if papain was used. [1]

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- (d) The kinetics of the decomposition of hydrogen peroxide in the presence of silver can be investigated by drawing portions of the solution at different time intervals, quenched and titrated with potassium manganate(VII).

A graph of volume of potassium manganate(VII) against time was then plotted.



- (i) Deduce the order of reaction with respect to $[\text{H}_2\text{O}_2]$. [2]

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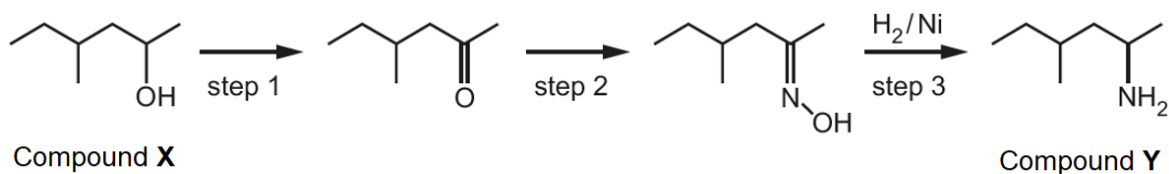
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- (ii) Suggest and explain the effect on the rate constant for the decomposition of hydrogen peroxide in the absence of silver. [1]

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- (e) Compound **Y** is a banned performance-enhancing stimulant. It may be prepared from compound **X** by the following three-step synthesis.



- (i) Name compound **X**. [1]

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- (ii) Suggest the type of reaction occurring in steps 1 and 3. [2]

step 1:

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step 3:

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- (iii) Compound **X** can react with hot Al_2O_3 to form a mixture of 3 isomers. Draw the structure of each isomer. [2]

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7 (a) Hydrazine, N_2H_4 , was first used as a component in rocket fuel in World War 2.

(i) Using the VSEPR theory, state and explain

- the shape of hydrazine around each N atom;
- the H-N-H bond angle.

[3]

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(ii) Hence, draw the structure of hydrazine, showing clearly the shape around one nitrogen atom. [1]

(iii) Like ammonia, hydrazine are bases.

Write an equation for the reaction between hydrazine and dilute hydrochloric acid.
Identify the conjugate acid-base pair in the reaction. [2]

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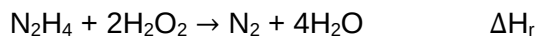
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- (b) When combined with hydrogen peroxide, the reaction produced nitrogen gas and water vapour, and released a large amount of energy that propelled the rocket to space.



- (i) Define the term *bond energy* of a covalent bond. [1]

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- (ii) Calculate the enthalpy change, ΔH_r , of the gaseous reaction between hydrazine and hydrogen peroxide. [2]

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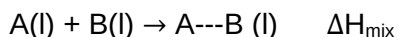
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- (c) When two pure liquids, A and B, which do not react, are mixed together, heat may be absorbed or released depending on the relative strength of intermolecular forces of attraction formed in the mixture.



--- refers to the intermolecular forces of attraction between liquids A and B.

Enthalpy change of mixing, ΔH_{mix} , is the enthalpy change when 1 mole each of pure A and pure B are added to form a binary A---B solution.

The enthalpy change of mixing of ethyl ethanoate and trichloromethane is determined experimentally in an insulated calorimeter and the following data is collected.

	ethyl ethanoate	trichloromethane
Molar mass / g mol^{-1}	88.0	119.5
Mass / g	8.80	—
Specific heat capacity / $\text{J g}^{-1} \text{K}^{-1}$	1.92	0.96
Initial temperature / $^{\circ}\text{C}$	31.2	31.2
Final temperature / $^{\circ}\text{C}$	40.7	40.7

- (i) Calculate the heat change of ethyl ethanoate during the mixing process. [1]

- (ii) The heat change for an equimolar of trichloromethane at the same initial temperature is 109 J.

Calculate the heat change of the mixture and hence the enthalpy change of mixing, ΔH_{mix} , of ethyl ethanoate and trichloromethane. [2]

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- (iii) With reference to structure and bonding, explain the significance of the sign of ΔH_{mix} in (b)(ii) and hence, predict the observation of mixing ethyl ethanoate and trichloromethane. [3]

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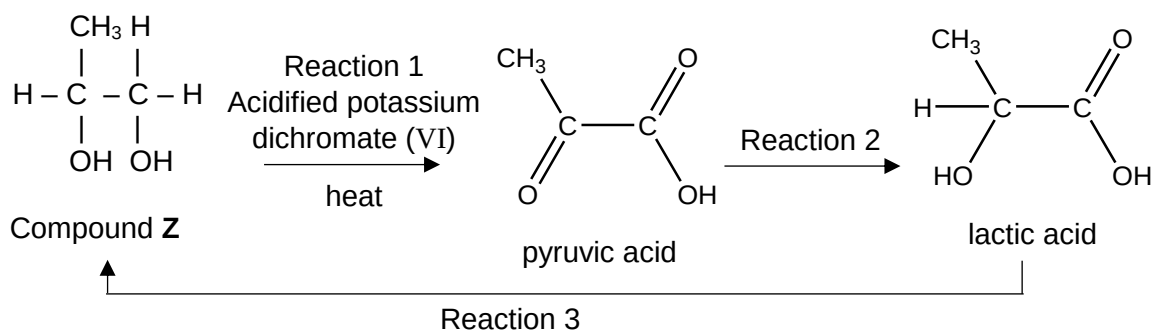
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- (d) Lactic acid can be formed from compound **Z** through the following reaction scheme.



- (i) State the reagents and conditions for reaction 2 and 3. [2]

Reaction 2

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Reaction 3

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- (ii) When Compound **Z** is warmed with acidified potassium dichromate (VI) with immediate distillation, another product is formed.

Draw the structure of the product formed.

[1]

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- (iii) Compound **Z** can undergo a series of reactions to form an alkene.

Describe, in terms of orbital overlap, the bonding of the two carbon atoms of the C=C bond in alkenes. A clearly labelled diagram may clarify your answer. [2]

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